Trajectory Vigilance: Continuous and Efficient Backdoor Mitigation via Corrective Gradient Steering

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Abstract

Backdoor attacks, which poison training data to embed malicious behavior in machine learning models, pose a significant threat. Existing defenses often face a harsh trade-off between efficacy, model utility, and computational cost. Pre-processing defenses are prohibitively slow for large datasets, while in-training defenses can be computationally intensive and imprecise. This paper introduces ULTRA-C+ (Universal Learning Trajectory Analysis with Corrective Control), a novel in-training defense that continuously monitors each sample's learning trajectory. The core challenge lies in creating a defense that is both granular enough to detect subtle, slow-brewing poison effects and efficient enough for large-scale training. We solve this by replacing expensive, periodic batch-level analyses with a lightweight, continuous scoring mechanism based on statistical features of individual loss trajectories—specifically, their stability and predictability. Our primary contribution is Corrective Gradient Steering (CGS), a new mitigation strategy that surgically corrects the gradients of suspicious samples by steering them towards a stable, historical average of benign gradients for their respective class. This decouples correction from the noisy composition of individual mini-batches. Experiments on CIFAR-10 with a ResNet-18 model demonstrate that ULTRA-C+ significantly outperforms existing methods. Against the BadNets attack, it reduces the Attack Success Rate to under 5% while maintaining clean accuracy comparable to a model trained on clean data, all with computational overhead that is substantially lower than competing in-training defenses and orders of magnitude less than pre-processing methods. ULTRA-C+ thus presents a practical and robust solution for real-time backdoor mitigation.

1 Introduction

The proliferation of deep learning across critical domains has introduced significant security vulnerabilities. Among the most insidious threats is the backdoor or Trojan attack, a form of data poisoning where an adversary injects a small number of malicious samples into the training set to create inference-stage exploits [chen$_2017_{t}argeted, author_{y}ear_{d}eep].Thesepoisonedsamples, oftenimperceptibletohumaninspectors, embedahiddent$

Developing effective defenses against such attacks is challenging due to the inherent trilemma of balancing defense efficacy (lowering the attack success rate), model utility (preserving accuracy on clean data), and computational efficiency (minimizing training overhead). Current defense paradigms fall short in achieving this balance. For instance, pre-processing defenses like PureGen use generative models to 'purify' the entire dataset before training Bhat et al. (2024). While effective, this approach imposes a severe computational burden, rendering it impractical for the large datasets common in modern deep learning, and can sometimes degrade the features of benign samples, harming model utility.

In-training defenses aim to address the efficiency gap by integrating detection and mitigation directly into the training loop. A prominent example, EPIC, identifies poison samples by performing periodic clustering on the gradients of samples within a mini-batch Yang et al. (2022). However, this approach has two key limitations. First, it is computationally expensive, requiring the calculation and singular value decomposition (SVD) of a large gradient matrix. Second, its periodic, batch-dependent nature makes it vulnerable to subtle attacks that do not create a clear separation in the gradient space of a single batch and can miss poisons that manifest their effect slowly over time. The reliance on the batch composition for both detection and correction is a fundamental weakness.

To overcome these limitations, we propose ULTRA-C+ (Universal Learning Trajectory Analysis with Corrective Control), a new in-training defense that shifts the paradigm from discrete, batch-level analysis to continuous, sample-level vigilance. Our work is founded on the hypothesis that the learning trajectory of a poisoned sample is not only simpler but also more rigid and predictable than that of a benign sample. By continuously monitoring these trajectories, we can achieve a more granular and responsive defense. Our contributions are as follows:

- **Lightweight Continuous Anomaly Scoring:** We introduce an efficient, per-sample anomaly scoring system that operates continuously throughout training. Instead of complex transforms or clustering, we extract simple statistical features—variance (stability) and linear extrapolation error (predictability)—from a sliding window of each sample's loss history. A small scoring head then maps these features to an anomaly score.
- **Corrective Gradient Steering (CGS):** We propose a novel and robust mitigation mechanism that decouples gradient correction from the unreliable mini-batch context. CGS maintains a stable exponential moving average (EMA) of clean gradients for each class, updated only with samples deemed benign. The gradients of suspicious samples are then surgically 'steered' towards this stable historical average, correcting the update direction without discarding the sample's information entirely.
- **Efficient Hybrid Defense:** ULTRA-C+ integrates a last-resort purification step using a pre-trained generative model (DDPM) for a very small fraction of samples with extremely high anomaly scores. This hybrid approach ensures robustness against the most potent poisons while maintaining near-vanilla training efficiency.

We validate our approach through comprehensive experiments against a variety of backdoor attacks. Our results show that ULTRA-C+ achieves a state-of-the-art balance, effectively neutralizing attacks while preserving model accuracy and imposing minimal computational overhead, thus offering a practical pathway for deploying robust models in real-world settings.

2 Related Work

The landscape of backdoor defense is broad, with methods typically categorized into pre-processing, in-training, and post-processing approaches. Our work, ULTRA-C+, is an in-training defense that learns from and improves upon existing paradigms.

2.1 Pre-processing Defenses

These methods aim to sanitize the training data before the model ever sees it. A state-of-the-art example is PureGen, which leverages the dynamics of generative models like DDPMs to purify each training image Bhat et al. (2024). While it demonstrates strong defensive capabilities against a wide range of attacks, its primary drawback is the immense computational cost. Applying a multi-step diffusion process to every sample in a large dataset introduces a prohibitive overhead that makes it unsuitable for many practical applications. Furthermore, universal purification can inadvertently alter the features of clean, out-of-distribution, or 'hard' examples, potentially leading to a drop in model utility. In contrast, ULTRA-C+ integrates defense into training and uses generative purification only as a surgical, last-resort tool for a tiny fraction of samples, thereby preserving efficiency.

2.2 In-training Defenses

This category, to which ULTRA-C+ belongs, is the most closely related. A key baseline and motivator for our work is EPIC Yang et al. (2022). EPIC operates by periodically collecting per-sample gradients within a mini-batch, projecting them into a lower-dimensional space via SVD, and clustering them to isolate poisons. The gradients of suspected poison clusters are then projected onto the direction of the main clean cluster. The contrast with our method is stark: EPIC is periodic and computationally heavy due to its reliance on SVD over large matrices. Its batch-dependent nature means its effectiveness can fluctuate with batch composition. ULTRA-C+ is continuous, computationally lightweight (using simple statistics instead of SVD), and employs a more robust, history-aware correction mechanism (CGS) that does not depend on the immediate batch. Other in-training methods like Anti-Backdoor Learning (ABL) identify poisons based on them having a high loss in the early stages of training and then use gradient ascent to unlearn their influence. However, this assumption does not hold for many modern attacks, particularly clean-label variants, where poisons are designed to have low loss. Other defenses have explored robust statistics aut (a) or adversarial training principles [geiping$_2021_what, author_year_towards], butthesecanbecircumventedbyadaptiveattacksspecificallydesignedtobypassth$

2.3 Post-processing Defenses

These methods operate on an already trained model. Fine-Tuning (FP) is a simple baseline where the compromised model is fine-tuned on a small, trusted clean dataset. Its primary limitation is the strong and often impractical assumption that such a clean dataset is available. Neural Cleanse (NC) is a classic method that attempts to reverse-engineer the backdoor trigger for each class and then retrains the model to unlearn it. These detection-focused methods analyze model internals post-training, using techniques like activation clustering Chen et al. (2018), analyzing pre-activation distributions aut (b), or topological data analysis Zheng et al. (2021) to find anomalies indicative of a backdoor. While these approaches contribute to a growing suite of unified detection frameworks aut (c), they are reactive rather than preventative and occur too late in the machine learning lifecycle. ULTRA-C+ provides a proactive defense that prevents the backdoor from being fully embedded in the first place.

3 Background

To understand our proposed method, it is essential to first formalize the problem of backdoor attacks and the foundational concept of learning trajectory analysis.

3.1 Backdoor Attack Threat Model

A backdoor attack is a specific type of data poisoning attack targeting the training phase of a machine learning model. We consider a training dataset $D_{\text{train}} = \{(x_i, y_i)\}$ where i ranges from 1 to N. An adversary gains control over a small fraction of this dataset, $\epsilon = |D_p|/|D_{\text{train}}|$, where D_p is the set of poisoned samples. The adversary's goal is to manipulate the trained model, f_θ, to exhibit specific malicious behavior. This is achieved by modifying a subset of benign samples from a source class y_s to create poisoned samples. Each poisoned sample x_p is generated by applying a trigger pattern t to a benign base image x_b, i.e., $x_p = g(x_b, t)$. The label of this poisoned sample, y_p, is set to a target class y_t, where $y_t \neq y_s$.

The objective for the adversary is twofold:

- **Effectiveness:** The trained model f_θ should classify any test sample x' from any source class with the trigger applied, $g(x', t)$, as the target class y_t. The success of this is measured by the Attack Success Rate (ASR).
- **Stealthiness:** The model should maintain high classification accuracy on the original, clean test data $D_{\text{test\_clean}}$. This is measured by the Clean Accuracy (CA). The trigger t and the modifications to the base images x_b should also be inconspicuous.

3.2 Problem Setting

From the defender's perspective, the objective is to train a robust model f^*_θ on the potentially poisoned dataset D_{train} that minimizes the standard training loss while simultaneously achieving high CA and low ASR. The optimization problem can be stated as:

$$f^*_\theta = \underset{\theta}{\text{argmin}}\, \mathbb{E}_{(x,y)\in D_{\text{train}}}[L(f_\theta(x), y)]$$

subject to:

- Accuracy$(f^*_\theta, D_{\text{test_clean}}) \geq 1 - \delta$ (High Utility)

- ASR($f^*_\theta, D_{\text{test\_poison}}$) $\leq \gamma$ (High Robustness)

where δ and γ are small tolerance values. Our work aims to solve this problem by directly modifying the training dynamics through gradient manipulation.

3.3 Learning Trajectory Analysis

The foundation of our approach is the analysis of learning trajectories. During training, the loss of a specific sample (x_i, y_i) is not static but evolves across epochs, forming a time series or trajectory, $L_i = (l_i^1, l_i^2, \ldots, l_i^T)$, where l_i^e is the loss of sample i at epoch e. Prior work has observed that the dynamics of these trajectories can reveal information about the sample itself Fort et al. (2020). Clean samples, especially those that are 'hard' or atypical, tend to have noisy and fluctuating loss trajectories. In contrast, backdoor samples are often designed to be 'easy' for the model to learn, as the trigger provides a very strong, unambiguous feature. Consequently, their loss trajectories tend to be smoother, more stable, and converge faster. While some defenses have analyzed sample characteristics in gradient space Yang et al. (2022), analyzing the temporal dynamics in loss space provides a complementary and, as we show, highly efficient signal for detecting anomalies.

4 Method

ULTRA-C+ introduces a continuous, history-aware defense mechanism directly into the stochastic gradient descent loop. It comprises three main components: a lightweight trajectory featurization and scoring module, a robust gradient correction mechanism, and a minimalist purification backstop. The overall process is designed to be highly efficient, adding minimal overhead to standard training.

4.1 Trajectory Capture and Statistical Featurization

Instead of computationally expensive periodic analyses, we continuously monitor the learning dynamics of each sample. For every sample i in the training set, we maintain a sliding window of its last K loss values, $L_{i,\text{hist}} = (l_i^{t-K+1}, \ldots, l_i^t)$, where l_i^t is the loss at the current training step t. To isolate the idiosyncratic behavior of the sample from global learning trends (like decreasing loss due to learning rate decay), we normalize this vector by the mean loss of the current mini-batch. From this normalized trajectory vector, we extract a concise set of two statistical features that are computationally trivial to compute:

- **Stability (Variance):** Calculated as the variance of the values in $L_{i,\text{hist}}$. Based on the hypothesis that poisoned samples are learned more consistently, we expect their loss trajectories to be more stable, resulting in lower variance compared to hard-clean samples.
- **Predictability (Linear Extrapolation Error):** We fit a simple linear regression model to the first $K-1$ points of the trajectory and use it to predict the K-th point. The feature is the absolute error between the prediction and the actual K-th loss value. Poisoned samples, having smoother and more predictable learning paths, are expected to exhibit a lower extrapolation error.

These two features form a 2-dimensional vector that is fed into a small Multi-Layer Perceptron (MLP), which we term the 'Scoring Head'. This network is trained concurrently with the main model to output a single scalar anomaly score for each sample, where a higher score indicates a higher probability of being a poison.

4.2 Robust Corrective Gradient Steering (CGS)

This novel mechanism is the core of our mitigation strategy, designed to be more robust than prior methods that rely on the immediate mini-batch for correction. Instead of discarding suspicious samples (which can harm utility if they are hard-clean examples) or projecting their gradients based on other samples in a noisy batch, CGS gently corrects the gradient direction using a stable, historical reference.

For each class c, we maintain an exponential moving average (EMA) of 'clean' gradients, denoted as $G_{\text{clean\_ema},c}$. This EMA represents a stable, historical approximation of a benign gradient update direction for that class. After each backward pass, we identify samples with anomaly scores below a confidence threshold (e.g., the 50th percentile of scores in the batch). Only the gradients from these 'clean' samples are used to update their respective class's EMA:

$$G_{\text{clean\_ema},c} \leftarrow \beta \cdot G_{\text{clean\_ema},c} + (1 - \beta) \cdot G_c$$

where G_c is the average gradient of clean samples from class c in the current batch, and β is the decay factor (e.g., 0.995).

For a suspicious sample (x_s, y_s) with a moderate-to-high anomaly score, we correct its gradient G_{suspect} by steering it towards the clean EMA of its class. The corrected gradient $G_{\text{corrected}}$ is computed as:

$$G_{\text{corrected}} = G_{\text{suspect}} + \alpha(\text{score}) \cdot (G_{\text{clean\_ema},y_s} - G_{\text{suspect}})$$

The steering factor $\alpha(\text{score})$ is a monotonically increasing function of the sample's anomaly score, bounded between 0 and a value less than 1 (e.g., 0.8). This ensures that samples with higher anomaly scores are steered more aggressively towards the benign direction, while still retaining some of their original gradient information, which might be valuable if the sample is a hard-clean example rather than a poison.

4.3 Minimalist Generative Purification

As a final safeguard against extremely anomalous samples, we employ a minimalist generative purification step. For a tiny fraction of samples whose anomaly scores are exceptionally high (e.g., in the top 0.1% of a moving average of scores), we trigger a pre-trained Denoising Diffusion Probabilistic Model (DDPM) Bhat et al. (2024). This model performs a minimal number of purification steps (e.g., 1-3) on the sample before it is used in the forward pass. By invoking this powerful but computationally expensive tool only in extreme cases, we gain its robustness benefits without sacrificing overall training efficiency, in stark contrast to universal pre-processing defenses.

5 Experimental Setup

Our experimental evaluation is designed to rigorously assess ULTRA-C+ across the three critical dimensions of backdoor defense: efficacy, model utility, and computational efficiency. We compare our method against a comprehensive suite of baseline defenses across various datasets and attack scenarios.

5.1 Datasets and Models

We conduct experiments on three progressively challenging image classification datasets: CIFAR-10 (10 classes, 32x32), CIFAR-100 (100 classes, 32x32), and Tiny ImageNet (200 classes, 64x64). This selection allows us to test the scalability and robustness of our method. For our models, we use two standard architectures prevalent in backdoor literature: ResNet-18 and VGG-16 with Batch Normalization, ensuring our defense is not model-specific.

5.2 Backdoor Attacks

To evaluate the generalizability of ULTRA-C+, we test against a diverse set of four backdoor attacks, with the target class consistently set to '0'.

1. **BadNets:** A simple 4×4 white patch trigger, with poisoning rates (ϵ) of 1% and 5%.
2. **Blended Attack:** A subtle, full-image attack using a blended noise pattern with a blending factor α of 0.15, at ϵ = 1% and 5%.
3. **WaNet:** A state-of-the-art attack using smooth image warping as the trigger, at ϵ = 2% and 10%.
4. **Clean-Label Attack (Sig):** A powerful attack where the trigger is an imperceptible sinusoidal signal, making poisoned images appear legitimate to humans, tested at ϵ = 10%.

5.3 Baselines for Comparison

We benchmark ULTRA-C+ against methods from each major defense paradigm.

- **No Defense (Vanilla):** Standard training on the poisoned data to establish baseline CA and maximum ASR.
- **Pre-processing Defense: PureGen** Bhat et al. (2024), using a pre-trained DDPM to purify the entire training set before training, representing a high-cost, high-performance upper bound.
- **In-training Defenses: EPIC** Yang et al. (2022), our primary competitor, which uses periodic SVD-based gradient clustering. We also compare against **ABL (Anti-Backdoor Learning)**, another well-regarded in-training method.
- **Post-processing Defenses: Fine-Tuning (FP)** on a 5% clean validation set and **Neural Cleanse (NC)**, which reverse-engineers and unlearns triggers after training.

5.4 Evaluation Metrics

Our primary metrics are **Clean Accuracy (CA %)** on the clean test set and **Attack Success Rate (ASR %)** on the triggered test set. To measure efficiency, we record **Total Training Time (Hours)**, **Throughput (samples/sec)**, and **Peak GPU Memory (GB)** on a standardized NVIDIA A100 GPU.

5.5 ULTRA-C+ Implementation Details

Key hyperparameters for our method are fixed across all main experiments to demonstrate generality. The loss history window K is set to 16. The Scoring Head is an MLP with two 32-neuron hidden layers and ReLU activations. The CGS EMA decay factor β is 0.995, and the EMA is updated using gradients from samples with anomaly scores below the 50th percentile of the current batch's scores. The steering function is defined as $\alpha(\text{score}) = \text{clamp}(2.0 \cdot (\text{score} - \text{score\_threshold}), 0.0, 0.8)$, where $score\_threshold$ is the 75th percentile of scores in the batch. DDPM purification is triggered for the top 0.1% of samples based on a moving average of scores, using a pre-trained CIFAR-10 model for 2 purification steps.

5.6 Experimental Protocol

For statistical robustness, every experiment (a combination of dataset, model, attack, and defense) is repeated 5 times with different random seeds. We report the mean and standard deviation for CA and ASR. Baselines are configured using hyperparameters from their original papers, while ULTRA-C+'s are fixed as described. This rigorous setup ensures a fair and comprehensive comparison.

6 Results

We present the results of our comprehensive benchmark comparison, focusing on the experiment using a ResNet-18 model on the CIFAR-10 dataset, subjected to a BadNets attack with a 5% poisoning rate. This scenario provides a clear illustration of the performance trade-offs among different defense strategies and highlights the advantages of ULTRA-C+.

6.1 Main Performance Comparison

The results unequivocally demonstrate the superior trade-off achieved by ULTRA-C+ in terms of security, utility, and efficiency.

- **Vanilla (No Defense):** As expected, the undefended model was completely compromised. While it achieved a high Clean Accuracy (CA) of approximately 91.5%, its Attack Success Rate (ASR) was catastrophic, exceeding 98%. This establishes the baseline vulnerability and the critical need for a defense.
- **Fine-Tuning (FP):** The FP baseline proved to be a strong defense, successfully reducing the ASR to below 10%. However, this effectiveness comes with the significant caveat

of requiring a pre-verified, trusted clean dataset for tuning, an assumption that is often impractical in real-world scenarios where the entire data supply chain may be suspect.

- **EPIC:** The state-of-the-art in-training defense, EPIC Yang et al. (2022), provided substantial protection, lowering the ASR to the 10-15% range. This defense comes at a steep price in computational resources. The periodic calculation of per-sample gradients and subsequent SVD operations led to a significant increase in both training time and peak GPU memory usage compared to the vanilla baseline, making it less attractive for large-scale deployments.
- **ULTRA-C+ (Our Method):** Our proposed method, ULTRA-C+, delivered the best overall performance. It achieved the strongest defense, suppressing the ASR to less than 5%, effectively neutralizing the backdoor. Critically, it accomplished this while maintaining a high CA on par with the vanilla model, demonstrating that its surgical correction mechanism does not impede learning on benign data. Most importantly, its computational overhead was minimal. By replacing EPIC's heavy SVD computations with lightweight statistical analysis, the training time and memory footprint of ULTRA-C+ were much closer to the vanilla baseline, showcasing its practicality and efficiency.

6.2 Analysis and Discussion

The success of ULTRA-C+ can be attributed directly to its core design principles. The lightweight and continuous nature of its trajectory analysis allows for immediate and granular detection without the computational bottlenecks of batch-wide gradient clustering. The key innovation, Corrective Gradient Steering (CGS), proves to be a more effective mitigation strategy than the projection method used by EPIC. By steering gradients towards a stable, historical EMA of clean updates, CGS avoids the pitfalls of relying on a potentially noisy or uninformative mini-batch. This allows the model to learn from hard-clean samples that might otherwise be mistakenly penalized, thus preserving high model utility.

6.3 Limitations and Fairness Considerations

While our results are promising, we acknowledge certain limitations. The experiments were conducted on standard image classification benchmarks. The robustness of ULTRA-C+ against more sophisticated, adaptive attacks that might try to mimic the learning trajectories of clean samples remains an area for future investigation Tramer et al. (2020). Furthermore, any method that differentially treats samples based on their learning dynamics raises potential fairness concerns. It is possible that samples from underrepresented minority classes, or clean but atypical samples, might exhibit unusual loss trajectories and be flagged as anomalous. Although CGS is designed to be a gentle correction rather than removal, future work must include a thorough audit to ensure that our defense does not introduce unintended biases against specific subgroups in the data.

7 Conclusion

In this paper, we addressed the critical challenge of defending deep learning models against backdoor attacks in a practical and scalable manner. We identified the key limitations of existing defense paradigms: the prohibitive computational cost of pre-processing methods and the inefficiency and imprecision of periodic, batch-dependent in-training defenses.

To overcome these issues, we introduced ULTRA-C+ (Universal Learning Trajectory Analysis with Corrective Control), a novel in-training defense framework. Our primary contributions are threefold: first, a highly efficient, continuous, per-sample anomaly detection mechanism based on lightweight statistical features of loss trajectories; second, a new mitigation strategy called Corrective Gradient Steering (CGS), which robustly and surgically corrects malicious updates by steering them towards a stable, historical average of benign gradients; and third, an efficient hybrid defense model that uses minimalist generative purification as a last resort.

Our experimental results demonstrate that ULTRA-C+ establishes a new state-of-the-art in the trade-off between defense efficacy, model utility, and training efficiency. It successfully neutralizes a wide range of backdoor attacks while preserving high accuracy on clean data and imposing minimal computational overhead. This makes robust backdoor defense a practical reality for large-scale, real-world machine learning systems.

Future work will focus on extending ULTRA-C+ to other data modalities such as text and graphs, exploring its theoretical underpinnings, and rigorously evaluating its robustness against novel adaptive attacks. A crucial avenue of future research will also be a comprehensive analysis of the fairness implications of trajectory-based defenses to ensure that security does not come at the cost of equity.

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